

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CSE)	Indian Institute of Technology, Jodhpur	8.84	2024–2028
Senior Secondary	CBSE Board, Shanti Asiatic School	91.4%	2024
Secondary	CBSE Board, Shanti Asiatic School	94.67%	2022

Experience

Undergraduate Researcher, CLARITY Lab, IIT Jodhpur under Dr. Hardik Jain (Assistant Professor) *Aug 2025–Apr 2026*

- Built [HandArM2](#), a gesture-controlled 3D model viewer using Python, MediaPipe and Ursina Engine with on-demand model import, enabling intuitive hand-driven navigation of complex 3D assets.
- Implemented exponential smoothing to reduce tracking jitter and improve interaction stability.
- **Technologies:** Python, MediaPipe, Ursina Engine, OpenCV, NumPy.

Projects

CubiCode / LocalCodeJudge

- CubiCode is a self-hosted LeetCode-style competitive coding platform; LocalCodeJudge is its Docker-based code execution engine that evaluates submissions in fully isolated containers.
- Built secure REST APIs to compile and execute code submissions across multiple languages with per-submission runtime, memory usage and verdict reporting.
- Developed a React coding workspace featuring a Monaco-based code editor, JWT authentication, MongoDB-backed problem management, and fully offline evaluation support.
- **Technologies:** React, Node.js, Express.js, MongoDB, Docker, JWT.

JaNET – Social Media Platform Analysis

- Modeled a social media network using weighted directed graphs representing users, follow relationships and community hierarchy for large-scale structural analysis.
- Implemented graph-based post recommendation and friend recommendation engines, along with chat connectivity analysis using BFS/DFS traversal.
- Designed efficient algorithms for interaction ranking and influence propagation across the user graph.
- **Technologies:** C++, Graph Algorithms, Data Structures.

Named Entity Recognition (NER) Analysis

- Analyzed and compared statistical and deep learning approaches for NER by studying contextual sequence modeling and entity classification.
- Built and benchmarked five NER architectures — HMM, MEMM, Vanilla RNN, CNN and BiLSTM , analysing trade-offs in accuracy, F1 and training efficiency.
- **Technologies:** Python, PyTorch, NumPy, Pandas, Scikit-learn.

Snail – Custom Linux Shell

- Developed a fully functional Unix-like shell supporting built-in and external commands, piping, I/O redirection, background execution and custom theming.
- Implemented persistent cross-session command history, an interactive `rm -i` confirmation flow, and a trash/restore system (`sendToDump/fileRestore`).
- **Technologies:** C, POSIX, Linux System Calls.

Key Courses Taken

Data Structures & Algorithms, Pattern Recognition & Machine Learning, Software Engineering, Operating Systems, Database Management Systems, Computer Networks, Artificial Intelligence

Technical Skills

- **Programming:** C, C++, Python, JavaScript
- **Tools & OS:** Git, Docker, Jupyter Notebook, Linux/POSIX, Windows
- **Libraries/Frameworks:** React, Node.js, Express.js, MongoDB, PyTorch, OpenCV, MediaPipe, NumPy, Pandas
- **Core:** Data Structures, Algorithms, Dynamic Programming, Graph Algorithms, Machine Learning

Achievements

- Ranked among the **top 35 students** in the CSE Department at IIT Jodhpur . 2024–Present
- Solved **320+ problems** on LeetCode with consistent focus on Dynamic Programming, Graphs, Binary Search and Sliding Window. Ongoing